

Stony Brook University
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Spring 2019

MATHEMATICS BACKGROUND QUIZ

NAME: _____

ID: _____

Instructions: This class requires a mathematical background at the level of MAT 125 or AMS 121. In the past, some students who had formally satisfied the requirement were not fully prepared, so this small quiz is meant to be a diagnostic tool to assess your readiness. Please try to answer as many questions as you can. No calculators are allowed.

GOOD LUCK!

1. Graph the following functions. Be as precise as you can, i.e. indicate intercepts, asymptotes, critical points and inflection points, if any.

(a) $y = -2x + 4$

(b) $y = \frac{2}{x}$

(c) $y = 16x - x^2$

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(d) $y = 4x - \frac{1}{x^2}$

2. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ for the functions below, where α and β are positive constants.

(a) $y = \alpha x^2 + \beta x$

(b) $y = \beta \ln(\alpha x)$

(c) $y = \alpha e^{\beta x}$

(d) $y = \beta z^2$ and $z = \alpha x + 5$

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(e) $y = \frac{\alpha x - \beta}{x - \alpha}$

3. Suppose a firm's total revenues depend on the amount produced (q) according to the function

$$R = 70q - q^2.$$

Total costs also depend on q :

$$C = q^2 + 30q + 100.$$

- (a) What level of output should the firm produce to maximize profits ($R - C$)? What will profits be?
- (b) Show that the second-order conditions for a maximum are satisfied at the output level found in part (a).
- (c) Does the solution that you calculated obey the "*Marginal revenue equals Marginal cost*" rule? Explain.